



PAPER – 2

ADVANCED FINANCIAL MANAGEMENT



QUESTIONS

Securitization

1. Grow More Ltd. an NBFC is in the need of funds and hence it sold its receivables to MAC Financial Corporation (MFC) for ₹ 100 million. MFC created a trust for this purpose called General Investment Trust (GIT) through which it issued securities carrying a different level of risk and return to the investors. Further, this structure also permits the GIT to reinvest surplus funds for short term as per their requirement.

MFC also appointed a third party, Safeguard Pvt. Ltd. (SPL) to collect the payment due from obligor(s) and passes it to GIT. It will also follow up with defaulting obligor and if required initiate appropriate legal action against them.

Based on above scenario, answer the following questions:

- I. The securitized instrument issued for ₹ 100 million by the GIT falls under category of
 - (a) Pass Through certificate (PTCs)
 - (b) Pay Through Security (PTS)
 - (c) Stripped Security
 - (d) Debt Fund.
- II. In the above scenario, the Originator is.....
 - (a) Grow More Ltd.
 - (b) MAC Financial Corporation (MFC)

- (c) General Investment Trust (GIT)
- (d) Safeguard Pvt. Ltd.
- III. In the above scenario, the General Investment Trust (GIT) is a/an.....
 - (a) Obligor
 - (b) Originator
 - (c) Special Purpose Vehicle (SPV)
 - (d) Receiving and Paying Agent (RPA)
- IV. In the above scenario, the Safeguard Pvt. Ltd. (SPL) is a/an.....
 - (a) Obligor
 - (b) Originator
 - (c) Special Purpose Vehicle (SPV)
 - (d) Receiving and Paying Agent (RPA)
- V. Which of the following statement holds true?
 - (a) When Yield to Maturity in market rises, prices of Principle Only (PO) Securities tend to rise.
 - (b) When Yield to Maturity in market rises, prices of Principle Only (PO) Securities tend to fall.
 - (c) When Yield to Maturity in market falls, prices of Principle Only (PO) Securities tend to fall.
 - (d) When Yield to Maturity in market falls, prices of Principle Only (PO) Securities remain the same.

Security Analysis

2. You are a financial analyst at a prominent investment firm and have been tasked with empirically verifying the weak form of Efficient Market Hypothesis (EMH) Theory for the XYZ Stock Index, a collection of diverse stocks. You decided to conduct three different tests to assess whether the stock market follows the principles of the weak form of EMH.

Test 1

For the past five years, you collected daily price changes of the stocks in the XYZ Stock Index. You calculated correlation coefficients for different lag periods and analyzed whether past price changes exhibit any significant correlation with future price changes. You considered price changes to be serially independent. The results indicated that most auto correlation coefficients are close to zero and statistically insignificant, suggesting those past price changes do not predict future price changes.

Test 2

You further investigated the randomness of price changes in the XYZ Stock Index. Analyzing the sequence of daily price changes, you count the number of runs where price changes are consistently positive or negative. Upon comparing the observed number of runs with the expected number based on randomness, you find that they align closely, supporting the idea that price changes follow a random pattern.

Test 3

To examine the efficacy of trading strategies based on historical price trends, you implemented a simple trading rule for the XYZ Stock Index. The rule involves buying when the price crosses a moving average of 5% threshold and selling when it crosses another 7% threshold. Over a period of testing, you computed the returns generated by the trading strategy. The results revealed that the returns are not consistently better than random chance, implying that past price trends do not reliably predict future price movements.

Conclusion:

After conducting the three tests the evidence supports the weak form of Efficient Market Theory for the XYZ Stock Index you concluded that past price trends do not reliably predict future price movements.

Based on the above information answer the following questions:

- I. Test 1 is
 - (a) Serial Correlation test
 - (b) Filter Rules test

- (c) Run test
- (d) Variance Ratio test
- II. Test 2 is
 - (a) Serial Correlation test
 - (b) Filter Rules test
 - (c) Run test
 - (d) Variance Ratio test
- III. Test 3 is
 - (a) Serial Correlation test
 - (b) Filter Rules test
 - (c) Run test
 - (d) Variance Ratio test.
- IV. The Filter Rule Test should not be applied for buy and hold strategy if.....
 - (a) the behavior of stock price changes is predictable.
 - (b) the behavior of stock price changes is dependent on past trends.
 - (c) the behavior of stock price changes is correlated.
 - (d) the behavior of stock price changes is random.
- V. Results of your studies support the.....
 - (a) Semi-strong EMH Theory
 - (b) Strong EMH Theory
 - (c) Random Walk Theory
 - (d) Markowitz Theory

Derivatives Analysis and Valuation

3. Mr. Shyam an investor is not sure about the expected price movement of the stock of Delta Corporation's share. His friend Adi advised him to go for option contracts if he wants to play in the market with limited risk. Adi

advised him to follow below mentioned Strategy.

- (1) Purchase one 3-month call option with a premium of ₹ 30 and an exercise price of ₹ 550.
- (2) Purchase one 3-month put option with a premium of ₹ 5 and an exercise price of ₹ 450.

Delta Corporation's stock is currently selling at ₹ 500.

Demonstrate the net pay off position of Mr. Shyam at the expiry of option after 3-months if the price of Delta Corporation's stock happens to be:

- (i) No change in price
- (ii) falls at ₹ 350
- (iii) rises to ₹ 600.

Assume the option lot size is 100.

4. The price of ACC stock on 31 December 2022 was ₹ 220 and the Futures price on the same stock on the same date, i.e., 31 December 2022 for March 2023 was ₹ 222. Other features of the Futures contract and related information are as follows:

Time to expiration	-	3 months (0.25 year)
Borrowing rate	-	15% p.a.
Annual Dividend on the stock	-	25% payable before 31.03. 2023
Face Value of the Stock	-	₹ 10

Advise the investor the course of action to be followed by him so as to earn Risk free income if he can sell the stock short at spot price.

Business Valuation

5. The following information is given for 3 companies that are identical except for their capital structure:

	Orange	Grape	Apple
Total invested capital	1,00,000	1,00,000	1,00,000
Debt/assets ratio	0.8	0.5	0.2
Pre-tax cost of debt	16%	13%	15%

Cost of equity	26%	22%	20%
Operating Income (EBIT)	25,000	25,000	25,000

The tax rate is uniform 35% in all cases.

Required:

- (i) Compute the Weighted Average Cost of Capital of each company.
- (ii) Tabulate the Economic Valued Added (EVA) of each company.
- (iii) If the industry PE ratio is 11x and stock prices of Orange, Grape and Apple are ₹ 14.30, ₹ 15.95 and ₹ 15.73 respectively then calculate the no. of shares issued by each company.
- (iv) Advise whether same Industry PE ratio can be used to calculate market price of share of each company.
- (v) Tabulate market Capitalisation for each of the Companies.

Foreign Exchange Exposure and Risk Management

6. JKL Ltd., an Indian company has an export exposure of JPY 10,000,000 receivable December 31, 2022. Japanese Yen (JPY) is not directly quoted against Indian Rupee.

The current spot rates are:

INR/US \$ = ₹ 82.22

JPY/US\$ = JPY 132.34

It is estimated that Japanese Yen will depreciate to 154 against US \$ and Indian Rupee to depreciate to ₹ 85 against US \$.

Forward rates as on date for 31st December 2022 are as follows:

INR/US \$ = ₹ 86.50

JPY/US\$ = JPY 140.35

Required:

- (i) Evaluate the expected loss based on estimated rates if the hedging is not done.
- (ii) Justify the decision to take forward cover even if actual rates on December 31, 2022 happens to be as follows:

INR/US \$ = ₹ 86.25

JPY/US\$ = JPY 140.85

Note: Make calculation of ₹ rate in ₹ upto 4 decimal points.

7. A UK based exporter exported goods to USA. The Invoice amount is \$ 7,00,000 and credit period is 3 months. Exchange rates in London are as follows: -

Spot Rate (\$/£) 1.5865 – 1.5905

3-month Forward Rate (\$/£) 1.6100 – 1.6140

Rates of interest in Money Market:

	Deposit	Loan
\$	7%	9%
£	5%	8%

Justify your stand to choose money market hedge (including steps) instead of Forward Contract.

Note: Make calculation upto 2 decimal points.

Advanced Capital Budgeting Decisions

8. X Ltd. is considering its new project with the following details:

Sr. No.	Particulars	Figures
1.	Initial capital cost	₹ 400 Cr.
2.	Annual unit sales	5 Cr.
3.	Selling price per unit	₹ 100
4.	Variable cost per unit	₹ 50
5.	Fixed costs per year	₹ 50 Cr.
6.	Discount Rate	6%

Required:

- (i) Tabulate the NPV of the project. Does it represent the actual outcome? Comment.

- (ii) Examine the impact of 2.5 percent adverse variance in each of the variables on the project's NPV. Decide which variable is having maximum effect?
- (iii) Critically analyse the Sensitivity analysis as method of incorporating risk in capital budgeting decisions.

Consider Life of the project as 3 years.

Interest Rate Risk Management

9. A Inc. and B Inc. intend to borrow \$ 200,000 and \$ 200,000 in ₹ respectively for a time horizon of one year. The prevalent interest rates are as follows:

Company	₹ Loan	\$ Loan
A Inc	5%	9%
B Inc	8%	10%

The prevalent exchange rate is \$ 1 = ₹ 120.

They entered in a currency swap under which it is agreed that B Inc will pay A Inc @ 1% over the ₹ Loan interest rate which the later will have to pay as a result of the agreed currency swap whereas A Inc will reimburse interest to B Inc only to the extent of 9%.

Keeping the exchange rate invariant, assess and analyse the opportunity gain or loss component of the ultimate outcome, resulting from the designed currency swap.

Mutual Fund

10. There are two Mutual Funds viz. D Mutual Fund Ltd. and K Mutual Fund Ltd. Each having close ended equity schemes.

NAV as on 31-12-2022 of equity schemes of D Mutual Fund Ltd. is ₹ 70.71 (consisting 99% equity and remaining cash balance) and that of K Mutual Fund Ltd. is ₹ 62.50 (consisting 96% equity and balance in cash).

Following is the other information:

Particular	Equity Schemes	
	D Mutual Fund Ltd.	K Mutual Fund Ltd.
Sharpe Ratio	2	3.3
Treynor Ratio	15	15
Standard deviation	11.25	5
Risk free rate of return (R_f)	7%	

There is no change in portfolios during the next month and annual average cost is ₹ 3 per unit for the schemes of both the Mutual Funds.

Required:

- Tabulate the expected NAVs of both the schemes if share market goes down by 5% within a month.
- Advise which mutual fund should an investor choose from the perspective of risk per unit of return.

Note: For calculation purpose, consider 12 months in a year and ignore number of days for a particular month.

Portfolio Management

- An investor has decided to invest ₹ 1,00,000 in the shares of two companies, namely, ABC and XYZ. The projections of returns from the shares of the two companies along with their probabilities are as follows:

Probability	ABC (%)	XYZ (%)
0.20	12	16
0.25	14	10
0.25	-7	28
0.30	28	-2

Required:

- Comment on return and risk of investment in individual shares.
- Compare the risk and return of these two shares with a Portfolio of these shares in equal proportions.

- (iii) Advise the proportion of each of the above shares to formulate a minimum risk portfolio.

Security Valuation

12. Mr. Amit is happy with the investment in a company as it is paying good dividend for the last few years. Last year it paid a dividend of ₹ 2 per share. The share is currently trading at ₹ 150 per share. He is of view that if he applies dividend discount model, the share is undervalued. As a financial expert examine his view that dividend discount model represents the fair value.

You being an expert is required to evaluate the market value of the share of the company.

Profit after tax of the company	₹ 290 crores
Equity capital of company	₹ 1,300 crores
Par value of share	₹ 40 each
Debt ratio of company (Debt/ Debt + Equity)	27%
Long run growth rate of the company	8%
Beta 0.1; risk free interest rate	8.7%
Market returns	10.3%
Capital expenditure per share	₹ 47
Depreciation per share	₹ 39
Change in Working capital	₹ 3.45 per share

Note: Round off figures (e.g. EPS etc.) upto 2 decimal points.

International Financial management

13. Mr. Vishwas, a friend of Mr. Pramod who is one of the Directors of Ashirwad Limited, is a citizen of Mauritius. His immediate family members including his parents, born in India are residing in India. He has many friends in different parts of India, due to which he happens to visit India on frequent basis. He along with Mr. Pramod evince interest in setting up business in India and formally incorporate a company to commence their operations. Accordingly, a company is called "Aerious Private Ltd." got incorporated in Mumbai.

To start with he received a business proposal from one of his friends Nimish a consultant. It is estimated that in equivalent terms the business shall require an initial investment of MUR 100 Million and thereafter MUR 2 Million each year will be needed as working capital fund.

He wished to evaluate whether the business proposal is viable or not. The information related to exchange rate and inflation rate is as follows:

Spot Rate for 1 Mauritian Dollar (MUR) = 1.88 Indian Rupee (INR)

The inflation in India is 6% and in Mauritius is 5%.

It is expected that this inflation rate will remain unchanged for the next 4 years.

INR 8 Crore out of initial investment shall be required for setting up a plant. The useful life of the plant is 4 years. At the end of 4th year estimated salvage value of this plant shall be INR 80 lakhs. Depreciation of the plant shall be charged on the basis of straight-line method.

40 % of the investment shall be through debt funds from Mauritius at the cost of 10% (post tax) while remaining funds shall be arranged by him and his friends. They expect a rate of return of 12% on their funds.

Expected revenues & costs (excluding depreciation) in real term are as under:

Year	1	2	3	4
Revenues (₹ Crore)	6.00	7.00	8.00	8.00
Costs (₹ Crore)	3.00	4.00	4.00	4.00

Assume that applicable tax rate in India is 30%. Since there is Double tax avoidance agreement between India and Mauritius, the company is not required to pay tax in Mauritius if tax has been paid in India.

The applicable inflation rates for revenues & costs are as follows:

Year	Revenues	Costs
1	10%	12%
2	9%	10%
3	8%	9%
4	7%	8%

He wants an expert opinion for the same investment proposal.

Demonstrate whether investment in this project is viable option or not.

- Note:** 1. Round off calculations upto 4 decimal points.
2. Show INR calculations in Crore and MUR calculations in Million.

Theoretical Questions

14. Succession planning is a good way for companies to ensure that businesses are fully prepared to promote and advance all employees—not just those who are at the management or executive levels. Explain the above statement.
15. In the current scenario of globalization and growth in information and communication technologies etc. the responsibilities of CFOs have been drastically expanded. Explain.